

1 Chapter 4: Documentation

2 **Reasons for changes:** The GMP/GDP Inspectors Working Group and the PIC/S Committee jointly
3 recommended that the current version of Chapter 4, on documentation, is revised to reflect changes in
4 regulatory and manufacturing environments. The revised guideline clarifies requirements and expecta-
5 tions from Regulatory Authorities with regards to documentation and takes into account related changes
6 for Annex 11 of the GMP Guide.

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7 PRINCIPLE

- 8 4.1. Documentation constitutes an essential part of the quality assurance system and is key to op-
9 erating in compliance with GMP requirements. The various types of documents and means
10 used should be fully understood and defined in the regulated user's pharmaceutical quality
11 system.
- 12 4.2. It should be determined by the regulated user which legal provisions apply to documentation
13 considering new technologies, hybrid solutions and services used.
- 14 4.3. Appropriate documentation practices should be applied with respect to the type of document
15 regardless of the applied technology or service used.
- 16 4.4. The present document was supplemented by requirements regarding new technologies, hybrid
17 solutions and new services, whereby a risk-based approach as element of a data governance
18 system is considered pivotal for scalability of integrity control measures.
- 19 4.5. Quality risk management principles should be applied to ensure that the pharmaceuti-
20 cal quality system includes sufficient instructional details. It should facilitate a common
21 understanding of the requirements. In addition to providing for recording of the various pro-
22 cesses and risk-based evaluation of any observations, it should demonstrate the ongoing appli-
23 cation of all requirements.
- 24 4.6. Suitable controls should be implemented using a risk-based approach to ensure the accuracy,
25 integrity, availability, and legibility of documents. Documents should be free from errors and
26 available in human readable form.
- 27 4.7. Documentation may exist in a variety of forms, including paper-based, electronic, or other
28 means (e.g. photography, imagery, video and audio recordings). The main objective of the
29 documentation system is to establish, control, monitor and record all activities which directly
30 or indirectly impact on all aspects of the quality and safety of medicinal products, using a risk-
31 based approach.
- 32 4.8. Whether documents are created, stored, and managed electronically, paper based, by other
33 means or through a hybrid system, they must meet the same GMP requirements for legibility,
34 accuracy, integrity, and completeness throughout the whole lifecycle. This also applies when
35 documentation is outsourced.
- 36 4.9. There are two primary types of documentation used to manage and demonstrate GMP compli-
37 ance:
 - 38 i. instructions (directions, requirements) and
 - 39 ii. records/reports.

40 DATA GOVERNANCE SYSTEMS

- 41 4.10. Regardless of how documents are created, handled, stored, and managed (i.e., using elec-
42 tronic, paper-based, or hybrid systems), the regulated user should establish a data governance
43 system integral to the pharmaceutical quality system to define, prioritise and communicate
44 their data integrity risk management activities. Arrangements for data governance should be
45 documented and reviewed regularly.
- 46 4.11. Regulated users should design and operate a data governance system which provides an ac-
47 ceptable state of control based on the risk assessment, which is documented with supporting
48 rationale. A data governance system should be consistent with the principles of quality risk
49 management.
- 50 4.12. To ensure integrity of data the governance system should cover the entire data lifecycle and
51 ensure controls commensurate with the principle of quality risk management. The data lifecy-
52 cle should refer to:
 - 53 i. Creation and recording of data.
 - 54 ii. Processing of (raw) data to reported (derived) data.

- iii. Verification of completeness, consistency and accuracy of all data (raw and derived data). For derived data the traceability which allows reconstruction of all data processing activities should be maintained.
 - iv. Decision making relying on data (or derived data).
 - v. Retaining, archiving and retrieval of data. To protect it from loss or unauthorised alteration it should be commensurate with the principles of quality risk management.
 - vi. Retirement or destruction of data at the end of the lifecycle in a controlled manner.
- 4.13. Data governance systems should rely on a risk management approach and consider:
- i. Data criticality (impact to decision making and product quality) and
 - ii. Data risk (opportunity for data alteration and deletion, and likelihood of detection / visibility of changes by the regulated user's routine review processes).
- 4.14. Data governance systems should rely on the incorporation of suitably designed systems, the use of technologies and data security measures, combined with specific expertise to ensure that data integrity is effectively controlled over the data lifecycle.
- 4.15. Data governance systems should address data ownership throughout the entire lifecycle.
- 4.16. Data governance systems should consider the design, operation and monitoring of processes and systems to comply with the principles of data integrity.
- 4.17. Data governance systems should consider risk mitigation. The effectiveness of risk mitigation measures should be reviewed regularly, regardless of whether they are temporary or permanent. Residual risks should be reviewed periodically and communicated to management.
- 4.18. Data governance systems should consider and ensure the periodic review of service provider's data management policies and risk control strategies intended to minimise potential risks to data integrity. The frequency of such reviews should be based on the criticality of the services provided, using risk management principles.

RISK MANAGEMENT

- 4.19. The regulated user should adopt a risk-based approach in documentation throughout the entire lifecycle of data, regardless of the technology, hybrid solution or service used and should demonstrate an understanding for data criticality, data risk and data quality.
- 4.20. Controls over the data lifecycle should be established which are commensurate with the principles of quality risk management. The depth of data governance and risk management activities should be justified and commensurate with the risks to product quality and patient safety.
- 4.21. Decisions on the extent of measures to ensure data integrity should be based on a documented rationale and documented risk assessment taking into consideration data criticality and data risk.
- 4.22. Irrespective of processes used to generate electronic data, they must be included in the requirements for the qualification or validation of the relevant computerised systems according to Annex 11.

GENERAL REQUIREMENTS FOR DOCUMENTATION

- 4.23. The pharmaceutical quality system should describe all documents required to ensure product quality and patient safety. Documents may be created, recorded, provided, approved, communicated, stored, and archived electronically, paper based or in a hybrid system. The reliance on electronic, paper-based or different means, hybrid solutions or hosted services in maintenance and retention of documentation requires the compliance with all EU GMP provisions including Annex 11 if decision making in manufacturing (e.g. batch release based on in-process controls and process analytical technologies) is supported by automatic validation scripts or artificial intelligence (Annex 22).
- 4.24. The accountability for the integrity of documents, records or (raw) data produced or processed

with artificial intelligence or any other automatic means (e.g. validation scripts) rests with the regulated user.

4.25. The support by any automatic means (e.g. validation scripts or artificial intelligence) should be included in a pharmaceutical quality system regardless of the service located on premise or as a hosted service. The records created electronically should enable a trend analysis of quality-critical data.

4.26. To ensure data integrity, data which is recorded or processed electronically should not be converted to or stored in a paper form unless it meets the requirements set out in section 13 “hybrid systems” or the conversion is validated or verified for accuracy.

MASTER DOCUMENTS

4.27. Specifically required master documents (not exhaustive list):

i. **Site Master File:** Refer to EU GMP Guidelines, Volume 4 “Explanatory Notes on the preparation of a Site Master File”.

ii. **Validation Master Plan:** A document describing the key elements of the site qualification and validation program. Master documents should be evaluated and reviewed on a regular basis.

iii. **Instructions** (directions, or requirements) type:

- **Specification:** Refer to glossary for definition

- **Manufacturing Formulae, Processing, Packaging and Testing Instruction:** Provide complete detail on all the starting materials, equipment, and computerised systems (if any) to be used and specify all processing, packaging, sampling, and testing instructions to ensure batch to batch consistency. In-process controls and process analytical technologies to be employed should be specified where relevant, together with acceptance criteria.

- **Procedures:** (Otherwise known as Standard Operating Procedures, or SOPs), documented set of instructions for performing and recording operations.

- **Protocol:** defined set of activities to provide instructions for performing and recording certain discreet operations.

- **Technical / Quality Agreement:** Written proof of agreement between contract givers and acceptors for outsourced activities.

iv. **Record/Report type:**

- **Record:** Provide evidence of various actions taken to demonstrate compliance with instructions, e.g. activities, events, investigations, and in the case of manufactured batches a history of each batch of product, including its distribution. Records include the raw data which is used to generate other records. For electronic records regulated users should define which data are to be used as raw data. At least, all data on which quality decisions are based should be defined as raw data.

Records can also exist as hybrid records which is a combination of paper records, electronic records or by other means. The completeness and integrity of records, including all relevant raw data and meta data should be ensured and protected based on risk.

- **Certificate of Analysis:** Provide a summary of testing results on samples of products or materials¹ together with the evaluation for compliance to a stated specification.

¹ Alternatively, the certification may be based, in-whole or in-part, on the assessment of real time data (summaries and exception reports) from batch related process analytical technology (PAT), parameters or metrics as per the approved marketing authorisation dossier.

- 145 - **Report:** Document the conduct of exercises, studies, assessments, projects or inves-
146 tigations, together with results, conclusions and recommendations.

147 **Specifications**

- 148 4.28. There should be approved and updated specifications for starting and packaging materials,
149 intermediate, bulk, finished products, process aids and other quality critical material, as ap-
150 plicable. Specifications should include all attributes which are relevant for product quality on
151 each stage of material or manufacture.

152 **Specifications for starting and packaging materials**

- 153 4.29. Specifications for starting and primary or printed packaging materials should include the prod-
154 uct and its reference code, if applicable:
- 155 i. A description of the materials, including:
 - 156 • The designated name and the internal code reference.
 - 157 • The reference, if any, to a pharmacopeial monograph.
 - 158 • The approved suppliers and, if applicable, the original producer of the material.
 - 159 • A specimen of printed materials.
 - 160 ii. Directions for sampling and testing.
 - 161 iii. Qualitative and quantitative requirements with acceptance limits.
 - 162 iv. Storage conditions and precautions.
 - 163 v. The maximum period of storage before re-examination.

164 **Specifications for intermediate and bulk products**

- 165 4.30. Specifications for intermediate and bulk products should be available for critical steps or if
166 these are purchased or dispatched. The specifications should be similar to specifications for
167 starting materials or for finished products, as appropriate.

168 **Specifications for finished products**

- 169 4.31. Specifications for finished products should include or provide reference to:
- 170 vi. The designated name of the product and the code reference where applicable.
 - 171 vii. The formula.
 - 172 viii. A description of the pharmaceutical form and package details.
 - 173 ix. Directions for sampling and testing.
 - 174 x. The qualitative and quantitative requirements, with the acceptance limits.
 - 175 xi. The storage conditions and any special handling precautions, where applicable.
 - 176 xii. The shelf-life.
- 177

178 **Manufacturing Formula and Processing Instructions**

- 179 4.32. Approved, written Manufacturing Formula and Processing Instructions should exist for each
180 product and batch size to be manufactured.

- 181 4.33. The Manufacturing Formula should include:

- 182 i. The name of the product, with a product reference code relating to its specification.
- 183 ii. A description of the pharmaceutical form, strength of the product and batch size.
- 184 iii. A list of all starting materials to be used, with the amount of each, described;

185 mention should be made of any substance that may disappear while processing, of
186 processing aids needed or any other material relevant for product quality.

187 iv. A statement of the expected final yield with the acceptable limits, and of relevant in-
188 termediate yields, where applicable.

189 4.34. The Processing Instructions should include:

190 i. A statement of the processing location and the principal equipment to be used.

191 ii. The methods, or reference to the methods, to be used for preparing the critical equip-
192 ment (e.g. cleaning, assembling, calibrating, sterilising).

193 iii. Checks that the equipment and workstation are clear of previous products, docu-
194 ments or materials not required for the planned process, and that equipment is clean
195 and suitable for use.

196 iv. Detailed stepwise processing instructions [e.g. checks on materials, pre-treatments,
197 sequence for adding materials, critical process parameters (time, temp etc)].

198 v. The instructions for any in-process controls with their limits.

199 vi. Where necessary, the requirements for bulk storage of the products; including the
200 container, labelling and special storage conditions where applicable.

201 vii. Any special precautions to be observed.

202 **Packaging Instructions**

203 4.35. Approved Packaging Instructions for each product, pack size and type should exist. These
204 should include, or have a reference to, the following:

205 i. Name of the product; including the batch number of bulk and finished product.

206 ii. Description of its pharmaceutical form, and strength where applicable.

207 iii. The pack size expressed in terms of the number, weight or volume of the product in
208 the final container.

209 iv. A complete list of all the packaging materials required, including quantities, sizes
210 and types, with the code or reference number relating to the specifications of each
211 packaging material.

212 v. Where appropriate, an example or reproduction of the relevant printed packaging
213 materials, and specimens indicating where to apply batch number references, and
214 shelf life of the product.

215 vi. Checks that the equipment and workstation are clear of previous products, docu-
216 ments or materials not required for the planned packaging operations (line clear-
217 ance), and that equipment is clean and suitable for use.

218 vii. Checks on functioning of any electronic code readers, label counters or similar de-
219 vices.

220 viii. Special precautions to be observed, including a careful examination of the area and
221 equipment in order to ascertain the line clearance before operations begin.

222 ix. A description of the packaging operation, including any significant subsidiary opera-
223 tions, and equipment to be used.

224 x. Details of in-process controls with instructions for sampling and acceptance limits.

225 **Batch Processing Record**

226 4.36. A Batch Processing Record should be kept for each batch processed. It should be based on

the relevant parts of the currently approved Manufacturing Formula and Processing Instructions, and should contain the following information:

- i. The name and batch number of the product.
- ii. Dates and times of commencement, of significant intermediate stages and of completion of production.
- iii. Identification (initials) of the operator(s) who performed each significant step of the process and, where appropriate, identification (initials) of the person who checked these operations.
- iv. The batch number and/or analytical control number as well as the quantities of each starting material weighed (including the batch number and amount of any recovered or reprocessed material added).
- v. Any relevant processing operation or event and major equipment used.
- vi. A record of the in-process controls and the initials of the person(s) carrying them out, and the results obtained.
- vii. The product yield obtained at different and pertinent stages of manufacture.
- viii. Notes on special problems including details, with signed authorisation for any deviation from the Manufacturing Formula and Processing Instructions.
- ix. Approval by the person responsible for the processing operations.

Note: Where a validated process is continuously monitored and controlled, then automatically generated reports may be limited to compliance summaries and exception/ out-of-specification (OOS) data reports.

With regards to decision making in manufacturing supported by automatic validation scripts or artificial intelligence refer to paragraph 4 of this document.

Batch Packaging Record

4.37. A Batch Packaging Record should be kept for each batch or part batch processed. It should be based on the relevant parts of the Packaging Instructions.

4.38. The batch packaging record should contain the following information:

- i. The name and batch number of the product.
- ii. The date(s) and times of the packaging operations.
- iii. Identification (initials) of the operator(s) who performed each significant step of the process and, where appropriate, the name of any person who checked these operations.
- iv. Records of checks for identity and conformity with the packaging instructions, including the results of in-process controls.
- v. Records of checks that the equipment and workstation are clear of previous products, documents or materials not required for the planned packaging operations (line clearance), that equipment is clean and suitable for use, and that any electronic code readers, label counters or similar devices are functioning as expected.
- vi. Details of the packaging operations carried out, including references to equipment and the packaging lines used.
- vii. Whenever possible, samples of printed packaging materials used, including specimens of the batch coding, expiry dating and any additional overprinting.
- viii. Notes on any special problems or unusual events including details, with signed

- 270 authorisation for any deviation from the Packaging Instructions.
- 271 ix. The quantities and reference number or identification of all printed packaging mate-
272 rials and bulk product issued, used, destroyed or returned to stock and the quantities
273 of obtained product, to provide for an adequate reconciliation. Where there are vali-
274 dated electronic controls in place during packaging there may be justification for not
275 including this information
- 276 x. Approval by the person responsible for the packaging operations.

277 **Receipt**

- 278 4.39. There should be written procedures and records for the receipt of each delivery of each starting
279 material, (including bulk, intermediate or finished goods), primary, secondary and printed
280 packaging materials and QC-samples. The records of the receipts should include:
- 281 i. The name of the material on the delivery notes and the containers.
- 282 ii. The "in-house" name and/or code of material (if different from a).
- 283 iii. Date of receipt.
- 284 iv. Supplier's name and, manufacturer's name.
- 285 v. Manufacturer's batch or reference number.
- 286 vi. Total quantity and number of containers received.
- 287 vii. The batch number assigned after receipt.
- 288 viii. Any relevant comment.
- 289 ix. If applicable, proof of verification that temperature during transportation were within
290 the approved limit.
- 291 4.40. There should be written procedures for the internal labelling, quarantine and storage of start-
292 ing materials, packaging materials, QC samples and other materials, as appropriate.

293 **Sampling**

- 294 4.41. There should be written procedures for sampling, which include the methods and equipment
295 to be used, the amounts to be taken and any precautions to be observed to avoid contamination
296 of the material or any deterioration in its quality (reference to EU GMP Guideline Volume 4,
297 Chapter 6 "Quality Control").

298 **Testing**

- 299 4.42. There should be written procedures for testing materials and products at different stages of
300 manufacture, describing the methods and equipment to be used. The tests performed should
301 be recorded (reference to EU GMP Guideline Volume 4, Chapter 6 "Quality Control").

302 **Other**

- 303 4.43. Written release and rejection procedures should be available for materials and products, and
304 in particular for the certification for sale of the finished product by the Qualified Person(s).
305 All records should be available to the Qualified Person at the time of the release decision. A
306 system should be in place to indicate special observations and any changes to critical data.
- 307 4.44. Records should be maintained for the distribution of each batch of a product in order to facil-
308 itate recall of any batch, if necessary.
- 309 4.45. There should be written policies, procedures, protocols, reports and the associated records of
310 actions taken, or conclusions reached, where appropriate, for GMP relevant actions, including

but not limited to the following examples:

- i. Validation and qualification of processes, equipment and systems.
 - ii. Equipment assembly and calibration.
 - iii. Data integrity.
 - iv. Technology transfer.
 - v. Maintenance, cleaning and sanitation.
 - vi. Personnel matters including signature lists, training in GMP and technical matters, clothing and hygiene and verification of the effectiveness of training.
 - vii. Environmental monitoring.
 - viii. Pest control.
 - ix. Complaints.
 - x. Recalls.
 - xi. Returns.
 - xii. Change control.
 - xiii. Investigations into deviations.
 - xiv. non-conformances e.g. out of specifications.
 - xv. Internal quality/GMP compliance audits.
 - xvi. Summaries of records where appropriate (e.g. product quality review).
 - xvii. Supplier audits.
- 4.46. Clear operating procedures should be available for major items of manufacturing and test equipment.
- 4.47. Logbooks should be kept for major or critical analytical testing, production equipment, and areas where product has been processed or handled. They should be used to record in chronological order, as appropriate, any use of the area, equipment/method, calibrations, maintenance, cleaning or repair operations, including the dates and identity of people who carried these operations out.
- 4.48. An inventory of documents within the pharmaceutical quality system should be maintained.

GENERATION AND CONTROL OF DOCUMENTATION

- 4.49. All types of documents (instructions and/or records) should be defined and adhered to regardless of the documentation technology, hybrid solution or service. The technology, hybrid solution or provided service needs to be understood regardless of complexity, should be adequately documented, and validated with risk-based controls in place. Relationships and control measures for master documents, official copies, data handling and records need to be defined for both hybrid and homogenous systems regardless of the type of service. Appropriate controls for documents should be implemented to ensure the completeness, integrity and legibility of the records throughout the lifecycle.
- 4.50. Documents should be designed, prepared, reviewed, and distributed in a controlled manner. They should comply with the relevant parts of Product Specification Files, Manufacturing and Marketing Authorisation dossiers, or dossiers of Investigational Medicinal Products, as appropriate. The reproduction of working documents from master documents should not allow any error or alteration to be introduced through

the reproduction process.

4.51. Documents should be regularly reviewed and kept up to date. Documents should be approved, signed, and dated by appropriate and authorised personnel. Documents should have unambiguous contents and be uniquely identifiable. The effective date should be defined.

4.52. Documents containing instructions should be laid out in an orderly fashion and be easy to review. The style and language of documents should fit with their intended use. Standard operating procedures, work instructions and methods should be written in an imperative mandatory style by using predefined format. Data entry formats for completion of documents should be clearly defined. Written instructions may be supported with pictures, photos or videos. The documents containing the instructions should be easily accessible at the place where the described activities are carried out.

4.53. Instructions and procedures within the Quality Management System should be regularly reviewed and kept updated.

4.54. The issuance, revision, superseding and withdrawal of all documents should be controlled with maintenance of revision histories.

4.55. Hand-written instructions are discouraged. Where documents require the manual entry of data, sufficient space should be provided for such entries to ensure adequately clear and legible manual recording.

GOOD DOCUMENTATION PRACTICE

4.56. Good Documentation practices are key to ensuring data integrity, and a fundamental part of a well-designed pharmaceutical quality system.

4.57. Procedures outlining good documentation practices and arrangements for document control should be available within the pharmaceutical quality system. Good documentation practices should be implemented and enforced to ensure data integrity.

4.58. Data entries should be accurate, and made in clear, legible, indelible way. Recorded media should be durable throughout the retention period. If this is not feasible, then true copies should be generated. For this case a documented system should be in place to verify and record the integrity of the copy.

4.59. Records should be made or completed at the time each action is taken and in such a way that all GMP activities are traceable. It should be possible to identify the individual or the system that performed the task and when the task was performed.

4.60. Any alteration made to the entry on a document should be signed by the individual who made the change and dated; the alteration should permit the reading of the original information. Where appropriate, the reason for the alteration should be recorded.

4.61. Records need to be a truthful and consistent representation of facts to ensure the accuracy of information, including data that is used to make critical decisions about the quality of products.

4.62. Specific controls should be implemented to ensure the integrity of raw data and results recorded on paper. These may include, but are not limited to:

i. control over the issuance and use of loose paper sheets (blank forms) at the time of recording data.

ii. control over the issuance of bound, paginated notebooks.

iii. control over the issuance and reconciliation of sequentially numbered copies of

blank forms with authenticity controls.

- iv. Control that raw data is contemporaneously and accurately recorded by permanent means.

4.63. Basic data integrity principles (table 1) applicable to both paper and electronic systems (i.e. ALCOA ++):

Table 1: Data integrity principles

Attributes	Requirement
Attributable	It should be possible to identify the individual or computerised system that performed a recorded task and when the task was performed. This also applies to any changes made to records, such as corrections, deletions, and changes where it is important to know who made a change, when, and why.
Legible	All records should be legible – the information should be readable and unambiguous in order to be understandable and of use. This applies to all information that would be required to be considered complete, including all original records or entries. Where the ‘dynamic’ nature of electronic data (the ability to search, query, trend, etc.) is important to the content and meaning of the record, the ability to interact with the data using a suitable application is important to the ‘availability’ of the record.
Contemporaneous	The evidence of actions, events or decisions should be recorded as they take place. This documentation should serve as an accurate attestation of what was done, or what was decided and why, i.e. what influenced the decision at that time.
Original	The original record can be described as the first capture of information, whether recorded on paper (static) or electronically (usually dynamic, depending on the complexity of the system). Information that is originally captured in a dynamic state should remain available in that state.
Accurate	Records need to be a truthful representation of facts to be accurate. Ensuring records are accurate is achieved through many elements of a robust pharmaceutical quality system. This can be comprised of: • equipment related factors such as qualification, calibration, maintenance, and computer validation. • policies and procedures to control actions and behaviours, including data review procedures to verify adherence to procedural requirements.